

Real Numbers — MCQ Worksheet

Mathematics • Chapter 1 • 25 Questions, 1 mark each — Total: 25 marks

Instructions: Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.

Q1. The HCF of two numbers a and b is the:

- (A) Largest number that divides both
- (B) Smallest number
- (C) Difference of a and b
- (D) Sum of a and b

Q2. Euclid's division lemma states that for any two positive integers a, b there exist unique integers q, r such that $a = bq + r$ where:

- (A) $0 \leq r < b$
- (B) $r > b$
- (C) $r < 0$
- (D) $r = b$

Q3. HCF (96, 404) is:

- (A) 2
- (B) 4
- (C) 8
- (D) 16

Q4. $\text{LCM} \times \text{HCF}$ of two numbers a, b equals:

- (A) $a + b$
- (B) $a - b$
- (C) $a \times b$
- (D) a/b

Q5. By Fundamental Theorem of Arithmetic, every composite number can be factorised as a product of:

- (A) Two primes only
- (B) Primes (uniquely up to order)
- (C) Two even numbers
- (D) Three primes

Q6. The prime factorisation of 156 is:

- (A) $2^2 \times 3 \times 13$
- (B) $2 \times 3^2 \times 13$
- (C) $2^2 \times 39$
- (D) 2×78

Q7. LCM of 15 and 20 is:

- (A) 20
- (B) 60
- (C) 100
- (D) 300

Q8. HCF of 24 and 36 is:

- (A) 6
- (B) 12
- (C) 18
- (D) 24

Q9. $\sqrt{2}$ is:

- (A) A rational number
- (B) An irrational number
- (C) An integer
- (D) A natural number

Q10. $5 + \sqrt{3}$ is:

- (A) Rational
- (B) Irrational
- (C) Integer
- (D) Natural

Q11. A rational number p/q ($q \neq 0$) has a terminating decimal expansion if q (in lowest form) has prime factors:

- (A) Of the form $2^m \times 5^n$ only
- (B) Includes 3
- (C) Includes 7
- (D) Includes 11

Q12. The decimal expansion of $13/3125$ is:

- (A) Non-terminating recurring
- (B) Terminating
- (C) Non-terminating non-recurring
- (D) Irrational

Q13. The decimal expansion of $23/8$ is:

- (A) Terminating
- (B) Non-terminating recurring
- (C) Non-terminating non-recurring
- (D) Irrational

Q14. The decimal expansion of $17/30$ is:

- (A) Terminating
- (B) Non-terminating recurring
- (C) Irrational
- (D) Non-real

- Q15.** The product of two irrational numbers is:
- (A) Always rational (B) Always irrational
(C) Sometimes rational, sometimes irrational (D) Always zero
- Q16.** $3 \times \sqrt{7}$ is:
- (A) Rational (B) Irrational
(C) Integer (D) Natural
- Q17.** If $\text{HCF}(a, b) = 5$ and $\text{LCM}(a, b) = 200$, then $a \times b$ is:
- (A) 100 (B) 200
(C) 1000 (D) 500
- Q18.** $\text{LCM}(8, 9, 25)$ is:
- (A) 1800 (B) 1500
(C) 800 (D) 2250
- Q19.** The smallest composite number is:
- (A) 1 (B) 2
(C) 4 (D) 6
- Q20.** The smallest prime number is:
- (A) 1 (B) 2
(C) 3 (D) 5
- Q21.** The HCF of two consecutive integers is:
- (A) 0 (B) 1
(C) 2 (D) Their sum
- Q22.** The HCF of two consecutive even integers is:
- (A) 1 (B) 2
(C) 4 (D) Their product
- Q23.** If p is a prime and p divides a^2 , then p divides:
- (A) a (B) a^2
(C) $2a$ (D) Always 0
- Q24.** Numbers like 0.10110011100011110... are:
- (A) Rational (B) Irrational
(C) Integers (D) Natural
- Q25.** The exponent of 2 in the prime factorisation of 144 is:
- (A) 2 (B) 3
(C) 4 (D) 5